Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/01/2021
PI Name:	Dynan, William Ph.D.		
Project Title:	Effect of GCR Exposure on the Functional Status of the DNA Repair Machinery and Its Contribution to Cancer Risk		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30322-4218	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	07/01/2018	**End Date:**	06/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	May 2019 report: None ; May 2020 report: None. May 2021 report: none.		
COI Name (Institution):	Yu, David M.D., Ph.D. (Emory University)		
Grant/Contract No.:	80NSSC18K1116		
Performance Goal No.:			
Performance Goal Text:			
	Current predictions of the risk of Radiation Exposure Induced Death have large uncertainties. In part, this is because of poor understanding of factors that influence risk in the low dose range corresponding to anticipated human exposure. Modeling suggests that the dose response may be strongly nonlinear, with an increase in risk at low doses attributable to Non-Targeted Effects (NTEs) in cells not directly traversed by a radiation track. A model incorporating NTEs, fitted to mouse Harderian gland carcinogenesis data, predicts that Mars mission cancer risk may be two-fold greater than that attributable to targeted effects alone. These findings highlight the need for a better understanding of the mechanisms underlying the high Relative Biological Effectiveness (RBE) of Galactic Cosmic Radiation (GCR) for carcinogenesis, including its ability to elicit NTEs. The principal hypothesis for this proposal is that exposure to GCR, and particularly its high charge and energy (HZE)		

Task Description:	particle components, is associated with chronic DNA damage signaling leading to decline in the functional status of the DNA repair machinery, and that this is a key contributor to GCR carcinogenesis. Prior studies establish the scientific premise for this hypothesis and identify potential underlying mechanisms, one of which is depletion of cellular NAD+ pools as a consequence of persistent DNA damage, resulting in reduced activity of important NAD+-dependent enzymes. These include the DNA damage sensing and repair factor, PARP1, and protein deacylases known as sirtuins. Experiments will investigate the role of the NAD+/PARP1/sirtuin axis in experiments using both human bronchial epithelial cells and wild-type C57Bl/6 mice. The work will focus on lung cancer risk as an endpoint. Specific Aims are: Aim 1. To investigate the scope and mechanistic basis for a GCR exposure-related decline in the functional status of the repair machinery using a bronchial epithelial cell model. (Removed from Study) Aim 2. To test the hypothesis that GCR exposure leads to a chronic DNA damage response in mouse lung and whether this can be abated with a dietary countermeasure. We will investigate gender, dose, and radiation quality effects. Aim 3. To determine whether radiogenic tumors display a mutational signature indicative of a decline in the functional status of the DNA repair machinery as reflected in their mutational signature, thus establishing a functional link to tumorigenesis. Anticipated deliverables include research results to improve evaluation of sex dependence of radiogenic lung cancer; to allow for calculation of RBEs including mixed field RBEs for males and females, particularly at low doses and in wild-type backgrounds where direct measurements of RBE for radiation-induced cancers is infeasible or subject to large experimental uncertainties; and to perform an initial evaluation of a potential countermeasure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The nicotinamide riboside countermeasure may have anti-aging properties that would make it useful outside the context of galactic cosmic ray exposure. Results may bear on this question.
Task Progress:	Major activities in Year 3 included completion of all scheduled mouse sacrifices and tissue harvest. Tissue harvest was completed in two extended campaigns in the Fall of 2020 and Spring of 2021. On an intent-to-treat basis, the cohorts that were sacrificed in year 3 included 1484 mice. The work was impacted by the COVID-19 pandemic. There were various challenges including short staffing, limits on room occupancies, and supply chain concerns. At this time, tissue samples are being held pending the arrival of a new staff member who will replace two staff members who left during the pandemic. The new staff member, who will begin on July 6, 2021, will coordinate embedding, sectioning, and scoring of lung tissue, together with other analyses. The other major activity in year 3 was to explore potential biomarkers of persistent DNA damage and/or repair deficiency. We had previously collected lung and other tissues from cohorts of mice at 1 and 6 months post-irradiation (n=5 males and 5 females per treatment group, per timepoint). For each mouse, the left lung was embedded and sectioned, and the right lung was flash frozen for biochemical analysis. We had earlier performed immunoblotting to analyze the levels of poly(ADP)-ribosylated (PARylated) proteins in extracts prepared from the frozen lung tissue. PARylation is a transient post-translational modification that occurs when poly (ADP)-ribose polymerases bind to single- or double-strand breaks. We had earlier seen large differences between irradiated and control groups, suggestive of radiation quality-dependent persistent genomic stress response. To complement the biochemical studies of protein PARylation, we performed immunohistochemistry studies focused on a set of approximately 60 samples (n=10 per group) representing mice that received the same 0.75 Gy dose of different radiation types. We probed with anti-PAR antibody, performed colorimetric staining quantified, and attempted to correlate to the biochemical results. Despite extensive optimization, there were high and variable backgrounds in many of the samples from non-irradiated mice, which blunted the ability to detect differences between groups. However a 28-Si irradiated group did show significantly more staining than a gamma-ray irradiated group (P=0.008), consistent with a radiation quality effect. In addition to PAR staining, we performed pilot experiments to investigate other markers of DNA damage and oxidative stress. We stained for phospho-XRCC1 (a marker of single-strand DNA breaks) and 4-hydroxynonenal (a lipid peroxidation product arising from chronic oxidative stress). Results were similar to PAR, that is, with high and variable backgrounds and no striking differences between treatment groups. We believe that the technical issues may be addressable via the use of a different detection method (multispectral fluorescence imaging), using facilities that are again available with the full reopening of the university in summer 2021 We hope to perform genome-wide expression analyses as a complement to immunohistochemistry (IHC) studies. In preparation for this, we isolated RNA from the remaining flash-frozen lung samples. Some of the lung samples were entirely consumed in the earlier analyses, but we were able to obtain sufficient material from 50 (of 80) mice Discussion: As the project moves into its fourth and final year, a first priority will be to complete the embedding, sectioning, and histologic examination of the 18-month lung samples. A second priority will be additional IHC analysis, using multispectral fluorescence imaging to quantify PAR, phospho-XRCC1, and potentially other markers. A third priority, as funds permit, will be genome wide expression analyses to complement the results of IHC. We have initiated a discussion with the NASA GeneLab whether they can perform RNA-Seq analyses, with the advantage that adoption of a common platform and methods would facilitate direct comparison with other NASA data sets. Results of these studies will potentially be informative with respect to dose response, radiation quality effects, sex differences, and underlying biology. Finally, if the number of high-quality tumor samples is sufficient, we will perform the tumor DNA sequencing and mutational analysis as envisioned in Aim 3. The results may be informative with respect to underlying biology of tumorigenesis.

Bibliography Type:	Description: (Last Updated: 08/06/2025)
Articles in Peer-reviewed Journals	Li Z, Jella KK, Jaafar L, Moreno CS, Dynan WS. "Characterization of exosome release and extracellular vesicle-associated miRNAs for human bronchial epithelial cells irradiated with high charge and energy ions." Life Sci Space Res (Amst). 2021 Feb;28:11-7. https://doi.org/10.1016/j.lssr.2020.11.001 ; PMID: 33612174 , Feb-2021
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